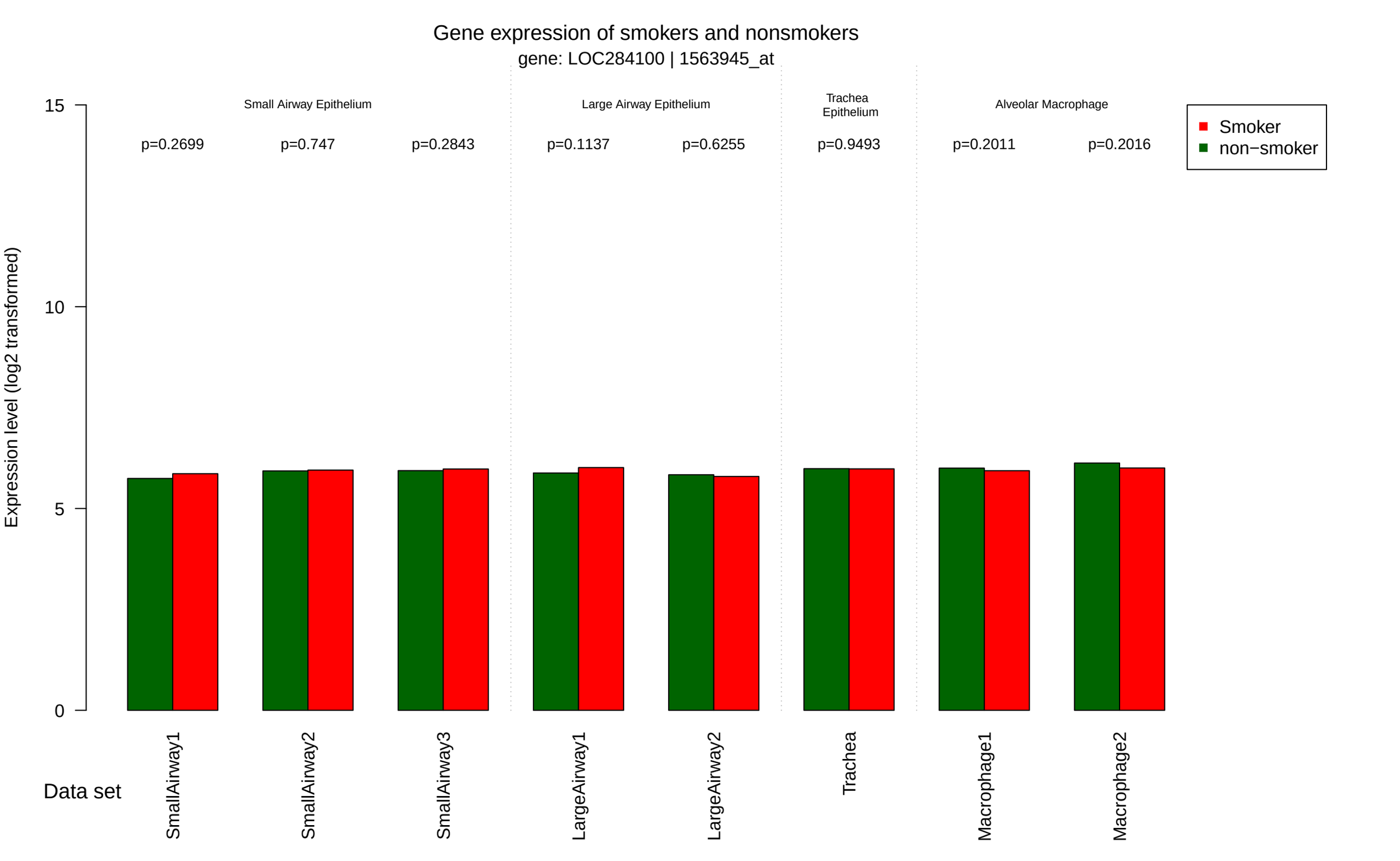
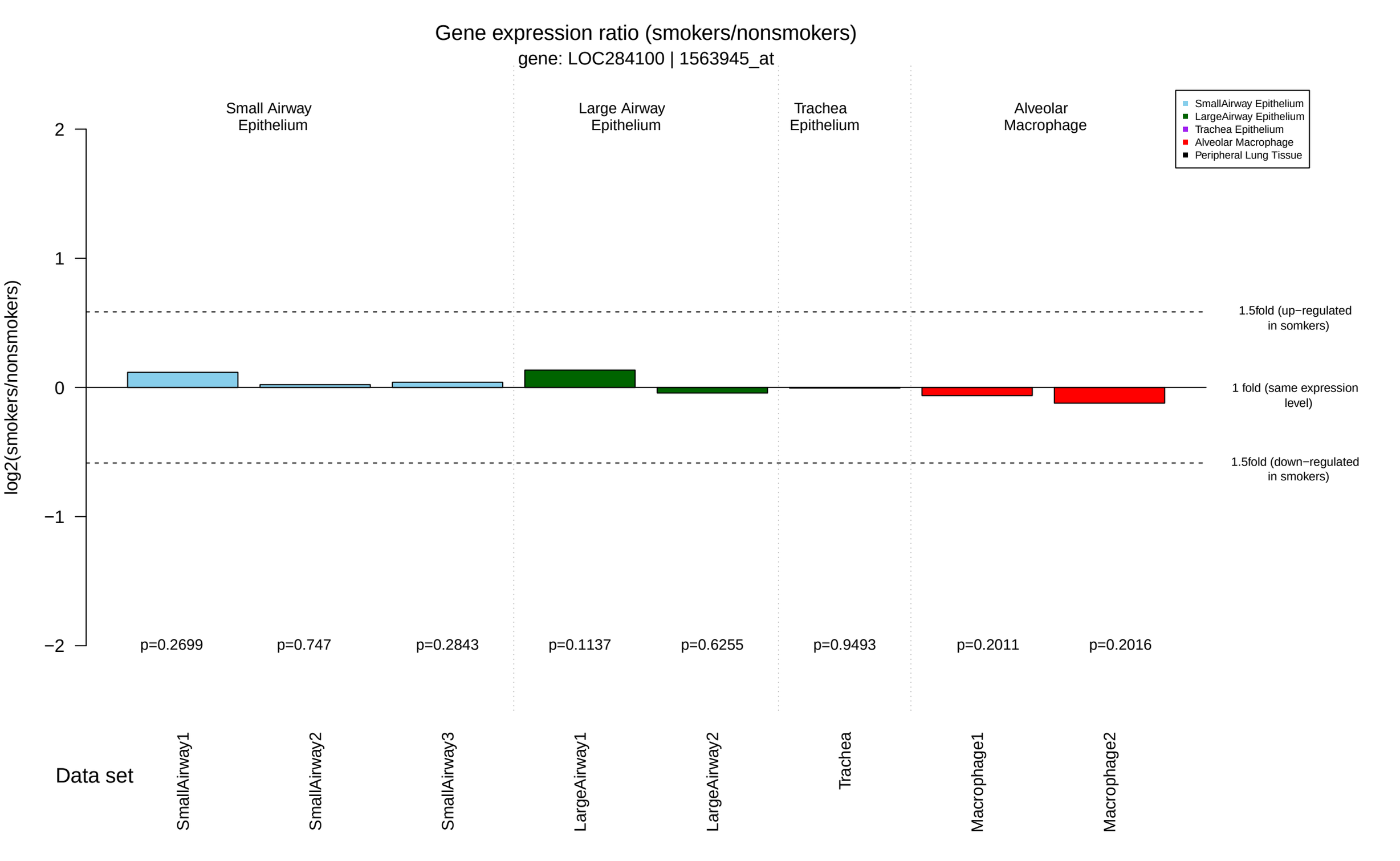
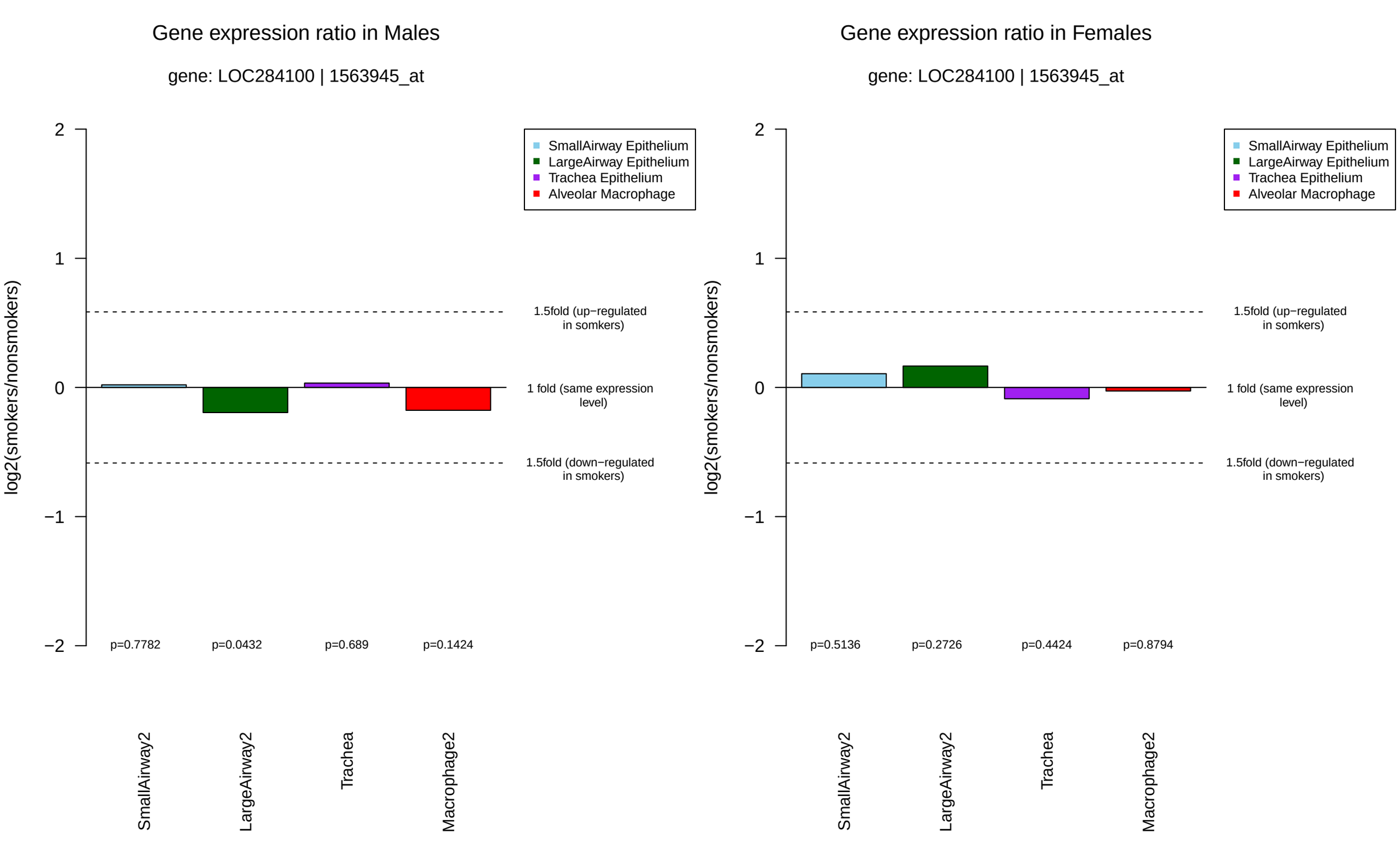
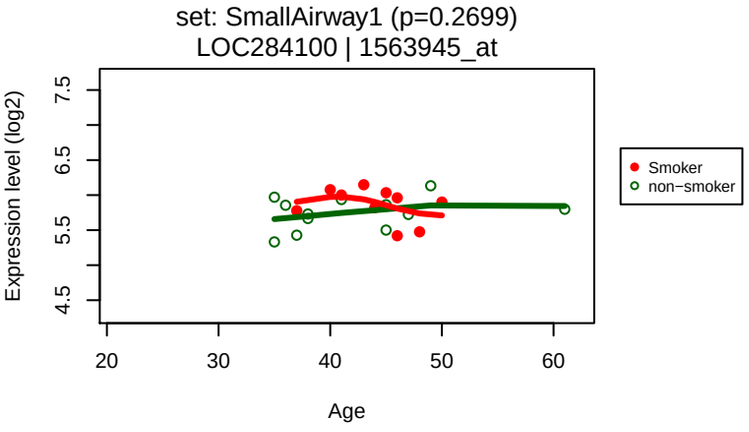
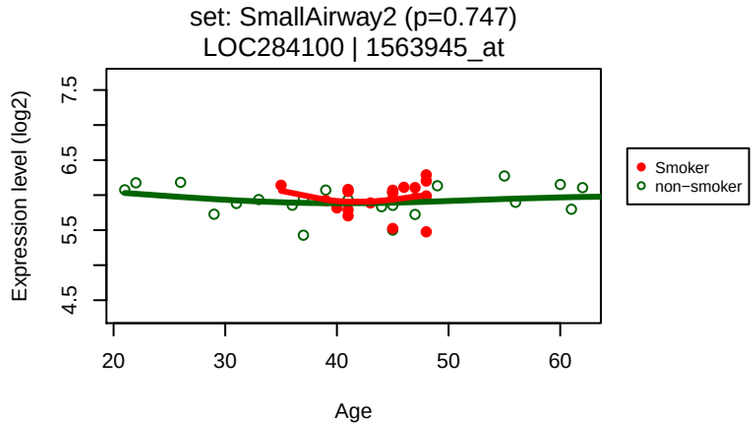
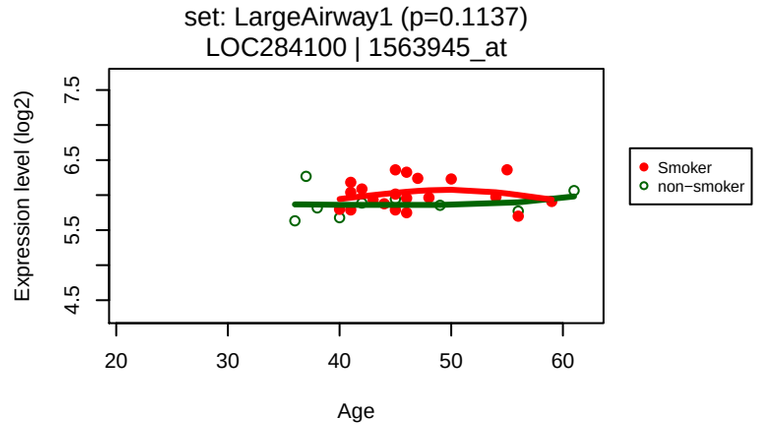

Gene expression of smokers and nonsmokers
gene: LOC284100 | 1563945_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2699
p=0.747
p=0.2843
p=0.1137
p=0.6255
p=0.9493
p=0.2011
p=0.2016
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

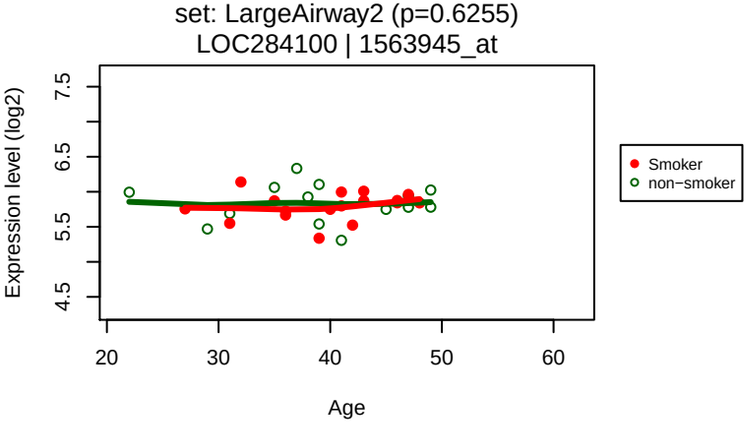

Gene expression ratio (smokers/nonsmokers)
gene: LOC284100 | 1563945_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2699
p=0.747
p=0.2843
p=0.1137
p=0.6255
p=0.9493
p=0.2011
p=0.2016
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

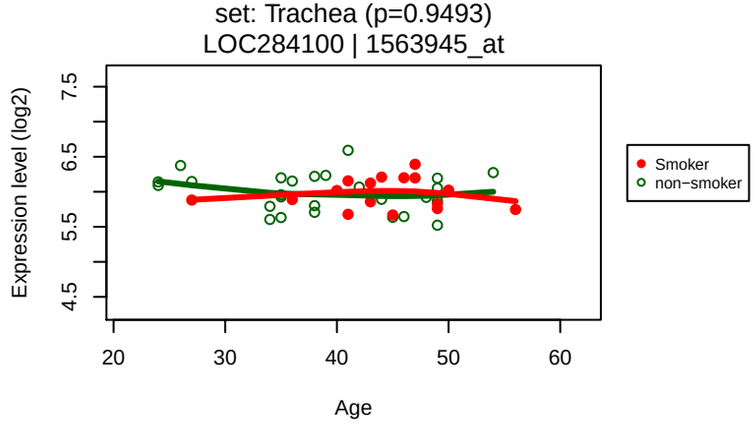

Gene expression ratio in Males
gene: LOC284100 | 1563945_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7782
p=0.0432
p=0.689
p=0.1424
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LOC284100 | 1563945_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5136
p=0.2726
p=0.4424
p=0.8794
SmallAirway2
LargeAirway2
Trachea
Macrophage2

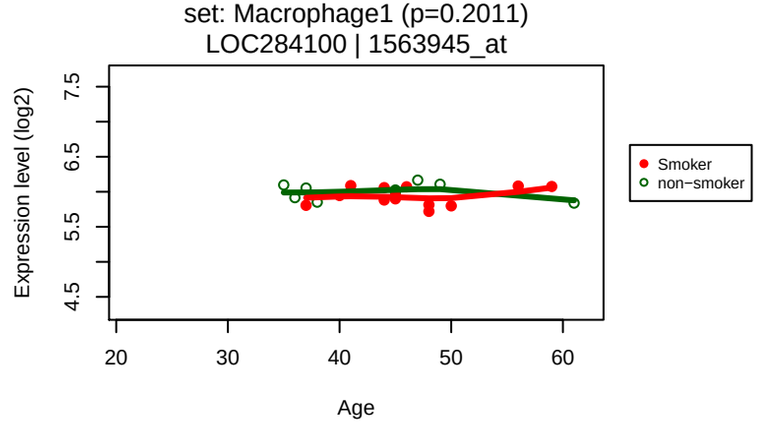

set: SmallAirway1 (p=0.2699)
LOC284100 | 1563945_at
Expression level (log2)
4.5
5.5
6.5
7.5
20
30
40
50
60
Age
Smoker
non−smoker

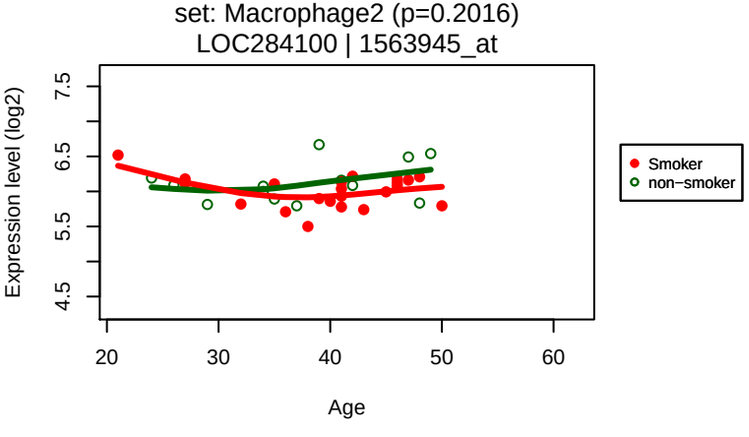

set: SmallAirway2 (p=0.747)
LOC284100 | 1563945_at
Expression level (log2)
4.5
5.5
6.5
7.5
20
30
40
50
60
Age
Smoker
non−smoker

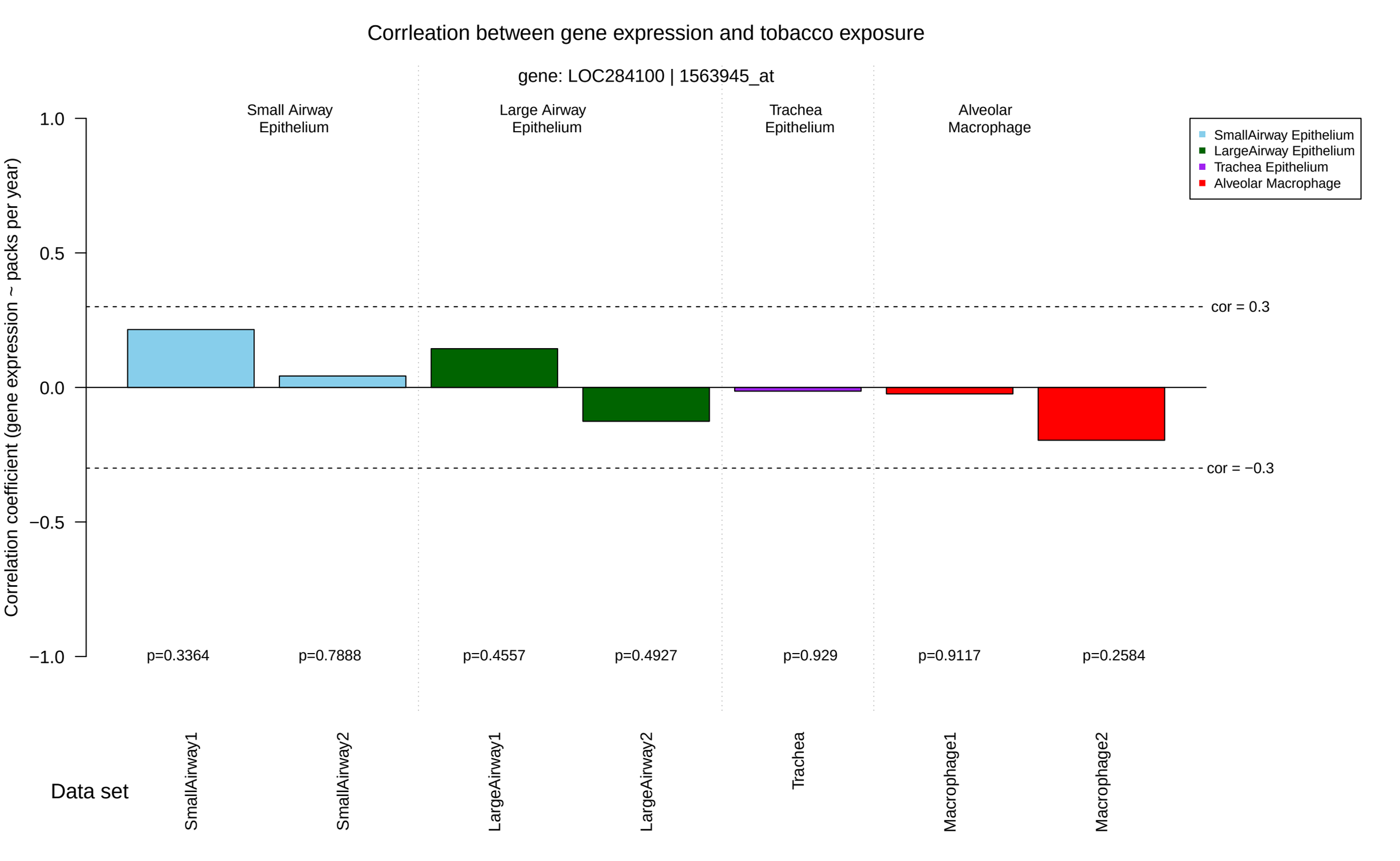

set: LargeAirway1 (p=0.1137)
LOC284100 | 1563945_at
Expression level (log2)
4.5
5.5
6.5
7.5
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6255)
LOC284100 | 1563945_at
Expression level (log2)
4.5
5.5
6.5
7.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.9493)
LOC284100 | 1563945_at
Expression level (log2)
4.5
5.5
6.5
7.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2011)
LOC284100 | 1563945_at
Expression level (log2)
4.5
5.5
6.5
7.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2016)
LOC284100 | 1563945_at
Expression level (log2)
4.5
5.5
6.5
7.5
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LOC284100 | 1563945_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.3364
p=0.7888
p=0.4557
p=0.4927
p=0.929
p=0.9117
p=0.2584
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2